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Section 1 Philosophical Context

The vast majority of human life, approximately 90 percent, is spent within indoor environments (Bennett et al., 2017), highlighting the critical importance of built spaces in shaping human experience. This reality underscores the significant influence that architectural design can exert on inhabitants' behavioral patterns and psychological responses (Baum & Valins, 2024). The extensive body of knowledge examining the relationship between humans and the built environment has been widely acknowledged and studied across multiple scientific disciplines such as psychology, neuroscience, sociology, and anthropology (Gifford, 2014; Karolis et al., 2019; St-Jean et al., 2022; Wang et al., 2022). Such approaches, however, have failed to address certain behaviors with a broader lens, as architecture inadequately incorporates insights from the field of behavioral science (Eren, 2017; Heimsath, 1978; Lang, 1988). Previous studies in behavioral science have not sufficiently examined the architectural design practice/profession itself (Lang, 1988), particularly with regard to sustainable design approaches at a systems level.

There is a critical need to shift the focus from buildings themselves to the design practices that create them (Schroeder, 2018). The persistent performance gaps in sustainable buildings and mismatches between design intentions and actual use suggest fundamental problems in how architectural practice conceptualizes and addresses human behavior. My study aims to address these issues through an interdisciplinary approach that views sustainable design elements as boundary objects (Leigh Star, 2010) at the intersection of architectural practice and behavioral science (Hong et al., 2017; Xu et al., 2023).

1.1 Research Statement

This research addresses critical disconnects between architectural design intentions and occupant experiences by examining architectural practice through a behavioral design systems lens. Despite extensive knowledge in both architecture and behavioral science/design, significant integration gaps persist at multiple levels: in understanding the complex system of architectural practice itself, in explaining performance discrepancies in sustainable buildings, and in aligning design intentions with actual use patterns. By developing a comprehensive methodological framework that leverages behavioral systems science as a theoretical foundation, this study explores the complex interactions between designed spaces and human behavior qualitatively, beyond conventional quantitative measures (Uddin et al., 2021). The research aims to systematically translate behavioral science principles into architectural design strategies that can more effectively encourage sustainable behavior in building occupants while analyzing the architectural practice itself (Hong et al., 2017; Maier & Cash, 2022).

1.2 Research Questions

- 1) How can architectural practice use a behavioral systems lens to inform more sustainable built environment/ architectural design practice?
 - a) How does the framing of 'sustainability' impact professional practice?
 - b) How do incentive structures shape behavioral responses toward sustainability in the context of architectural practice?
 - c) In what ways does 'choice infrastructure' of architectural practice itself shape decisions about design and development?

- 2) How can behavioral insight in the context of architectural practice help architects design built-environment-as a system to encourage sustainable behavior in inhabitants?

1.3 The Built Environment

Buildings are responsible for 35% of global energy consumption and almost 40% of total direct and indirect energy-related CO₂ emissions, effectively representing the largest energy-consuming asset in a city (Wang et al., 2017; Liu et al., 2022). Architecture has historically privileged hard sciences over soft sciences in its professional practice, creating a notable imbalance in its practice (Blibech, 2025). The discipline routinely incorporates engineering principles, materials science, physics, and mathematics to ensure structural integrity, environmental performance, and constructability (Bechthold & Weaver, 2017; Chen et al., 2022; Kurath, 2015). This emphasis on technical aspects stems from architecture's roots in engineering and construction, where measurable outcomes and predictable performance metrics dominate the discourse. Building codes, technical specifications, and computational modeling are all grounded in quantifiable data and empirical testing, form the backbone of architectural production and validation (Delzendeh et al., 2017; Feng et al., 2015). These hard science applications provide measurable outcomes and verifiable solutions that align with the practitioners' design. However, this rigid adherence to quantifiable metrics notable gap in our understanding of how buildings interact with their primary purpose: serving human occupants (Stazi et al., 2017; Yan et al., 2015; Yousefi et al., 2017). Contemporarily, a significant performance gap exists between predicted and actual building energy consumption. This discrepancy can be dramatic, with some cases showing actual energy use up to 300% higher than initial predictions (Delzendeh et al., 2017; Fabi et al., 2013; Yan et al., 2015). This substantial variance between expected and real-world energy performance represents a critical challenge in building design and energy efficiency assessment.

In the context of behavioral approaches, a number of studies have enhanced our understanding of how occupant behavior affects building energy consumption, such as building simulation modelling. Despite this they have not yet substantially improved the ability to predict occupants' energy behavior in buildings (Schweiker & Wagner, 2016; Stazi et al., 2017; Yan et al., 2015). A major research challenge remains in translating and integrating these research findings into building energy simulation tools to reduce the disparity between predicted and actual building energy consumption (Delzendeh et al., 2017; Heydarian et al., 2020).

Additionally, while architects had always considered basic human factors like ergonomics and circulation, the deeper behavioral aspects of space often receive less attention. The field of behavioral science would help understand and inform how spaces actually affect human behavior, interactions and understand the disconnect in architectural practice and real-world building usage. Despite architecture's ultimate purpose of creating environments for human use, the profession has developed fewer standardized methodologies for incorporating behavioral science theory and research into both the practice and field.

1.3.1 User Research

There has been a notable amount of research directly related to end-users in the context of architectural practice. For example, Ghisi et al. (2021) has embraced Human-Centric Design approaches, while others have incorporated frameworks like the Theory of Planned Behavior from

psychology (Heydarian et al., 2020). Evidence-Based Design has gained traction, as has Post-Occupancy Evaluation (POE), which Hay et al.(2018) and Eren (2017) highlight as crucial for addressing the performance gap between intended and actual building outcomes. These approaches have helped architects demonstrate the value of good design, providing evidence for investment in quality architecture and enabling decisions based on data rather than intuition alone.

Despite these advances, architectural practice hasn't fully utilized behavioral design's interdisciplinary potential. Architects often borrow isolated concepts from psychology or sociology but miss behavioral science's comprehensive approach to understanding human behavior across multiple analytical levels. While the profession acknowledges the need to move beyond technical metrics toward evaluating broader social outcomes like productivity, wellbeing, and community benefits, it lacks robust frameworks to do so. Many building occupants remain unaware of sustainable features in their environments, highlighting a critical gap (Li et al., 2024). The core challenge is that architecture has sampled from social sciences without adopting the holistic, theoretically grounded perspective that behavioral design offers for understanding the complex relationship between humans and their built environments.

1.3.2 Design Process

Research within the built environment has produced numerous frameworks for integrating sustainable practices into design, from technical assessment tools to the design process-oriented approaches (Feria & Amado, 2019; Stipo, 2015). However, a significant gap exists in understanding how external factors (client priorities, investors, budget constraints, regulations, market pressures, and stakeholder relationships etc.) fundamentally shape and often constrain implementation. This overemphasis on process-based solutions without considering the complex system of external influences has resulted in frameworks that appear comprehensive but fail to address real-world conditions determining whether sustainable design intentions translate into environmental performance.

The field's disproportionate focus on developing idealized sustainable design frameworks while neglecting powerful external factors represents a critical blind spot in architectural practice research. By treating sustainability primarily as a design process challenge rather than acknowledging how contextual forces fundamentally determine outcomes, researchers and practitioners have created an artificial separation between theory and practice. The disconnect between design frameworks and external forces may account for buildings' consistent underperformance against sustainability targets. To address this gap effectively, we must look beyond perfecting methodologies to critically examine how the entire system of client demands, financial constraints, regulatory environments, and market dynamics shapes outcomes from concept to construction.

1.3.3 Soft Sciences

Environmental psychology, neuroscience, psychology and other social sciences fields have provided essential frameworks for understanding human-environment interactions within the built environment. The evidence unequivocally demonstrates that during the rise of behavioral science, the field of architecture has historically been prescient in early attempts to create more human-centered design methodologies. Theories that focus on small-scale social systems, rather than

taking a holistic approach, include Behavior Setting Theory and Affordances (Barker, 1960; Brown et al., 2007; Gibson, Skinner, 1957; Zimmerman et al., 1991). A parallel systematic investigation by Heimsath's (1978) presents "*Behavioral Architecture*," prioritizing human needs through an "*inside out*" design approach that identifies activities, understands users, maps interactions, and creates supportive environments. This methodology emphasizes that architects with facility experience gain crucial behavioral insights, recognizing architecture's purpose extends beyond aesthetics to enhancing human activities and relationships. Notwithstanding, in contemporary architectural discourse, the field has not yet fully integrated behavioral science methodologies to inform either research or practice—presenting significant underexplored opportunities—while the potential contributions of social sciences are frequently overshadowed by technical and quantitative approaches that privilege hard data over qualitative insights that will be articulated more extensively in a subsequent section.

1.4 Behavioral Science

Behavioral and social sciences seek to identify, characterize, and explain social and behavioral phenomena through scientific principles and methods. Like all scientific disciplines, researchers in this field are driven by an innate curiosity and imaginative capacity to understand and theorize about human behavior and social dynamics (Gerstein et al., 1988). Through methodologically rigorous approaches, behavioral science builds and confirms theories about human conduct at both individual and collective levels, working to uncover the intricate factors that shape human behavior. Within this field, researchers place particular emphasis on studying dual-process theories, which examine how distinct cognitive mechanisms work in concert to influence our choices and actions (Hallsworth, 2023).

Behavioral science is commonly used in fields such as public policy, economics, healthcare, and organizational management, where principles of psychology and cognitive science are applied to understand and influence human decision-making processes (Altieri et al., 2021; Gintis, 2007; Straßheim, 2020; Thaler, 2018). These disciplines have embraced behavioral insights to develop effective interventions that account for how people actually behave rather than how they should theoretically behave. However, in architecture, the integration of behavioral science remains surprisingly limited despite the field's fundamental purpose of creating spaces for human habitation and activity (Eren, 2017). While post-occupancy evaluations and environmental psychology offer some intersection (Gifford, 2014; Hay et al., 2018; Ulrich et al., 2010), systematic application of behavioral insights throughout the architectural design process — and in particular, how architects themselves are affected by cognitive shortcomings and biases in the context of their work — is rarely observed (Klotz et al., 2018). This disconnect may stem from architecture's traditional emphasis on technical requirements, engineering and design intuition rather than research driven approaches to human behavior (Aburamadan & Trillo, 2020). Correspondingly, architectural education often prioritizes creative expression and technical proficiency over rigorous understanding of behavioral mechanisms, creating a professional culture where designers may recognize the importance of human factors but lack the methodological framework to systematically incorporate behavioral science into their practice (Lang, 1988). Thus, as the literature reveals, what remains notably absent is a robust framework that bridges architectural design with behavioral science and occupant behavior (Hong et al., 2017) — a critical gap that constitutes the central focus of my research inquiry.

Section 2 Empirical Context

Sustainability

Sustainability has become a pervasive topic in contemporary discourse both within and beyond architecture, evolving from a niche concern to a central consideration across disciplines, policies, and public consciousness (Caradonna, 2022). The urgency of climate change, resource depletion, and environmental degradation has elevated sustainability from an optional perspective to an essential framework for evaluating progress and development (Abbass et al., 2022; Mohy-ud-Din et al., 2025). Despite apparent consensus on the importance of sustainability in architecture, the field remains deeply contested, with practitioners and theorists divided over its interpretations, methodologies, and practical applications (Schroeder, 2018). The built environment has looked into the connection between pro-environmental impact and humans, including considerations of pro-environmental behavior, biophilic design, and environmental psychology (Gifford, 2014; Gifford et al., 2011; Heydarian et al., 2020; Pol, 2006).

However, the concept of sustainability is understood and employed differently by distinct disciplines and audiences. In architectural practice, sustainability applications are often formalized through standards and physical implementations through design approaches, materials/resources, and energy and certification systems (Jamoussi et al., 2022; Mi, 2024). In contrast, behavioral science often sees sustainability as a domain in which to influence specific “*end user*” human behaviors — e.g., nudging people to recycle or reduce energy consumption — based on insight into levers such as social norms, effort aversion, or habits (Klotz et al., 2019; Thaler, 2018). Pointing to a significant opportunity for greater integration between architectural and sustainable practice and behavioral insights to create built environments that more efficiently build, sustainably effective and support sustainable human behaviors.

Architectural design practices lack a rigorous approach to addressing the complexity and controversy inherently involved in sustainability (Schroeder, 2018). This conceptual breadth raises important questions about how sustainability is interpreted and operationalized across different domains. While the urgency of sustainable development is widely acknowledged, the translation of sustainability principles into practice reveals significant divergence in approach and emphasis between disciplines, often creating disconnects between theoretical frameworks and their practical implementation.

To comprehensively analyze sustainability within architectural practice, I adopt a multi-level approach examining how sustainable principles and behaviors manifest across three critical scales: individuals (architects themselves), practices (architectural firms), and disciplinary norms (architecture as a profession). This systematic examination allows for a nuanced understanding of how sustainability concepts operate differently yet interdependently across these levels, revealing both barriers to implementation and potential leverage points for intervention.

2.1 Individual (Architects)

2.1.1 Reputation/Incentive

Architects are driven by both external standards (e.g. LEED) and internal indicators like professional reputation (Question 1b). The importance of these motivational factors is heightened by evidence suggesting reputation serves as a powerful lever for behavioral change, especially

significant given architects' long-term impact on built environments. This perspective aligns with Sawyer's (2015) examination of architectural ego, which he attributes to the deeply personal nature of architectural work and defensive postures developed from constant criticism (relating to research question 1b on how incentive structures shape behavioral responses toward sustainability). While acknowledging some truth to this stereotype, Sawyer emphasizes that architects should prioritize problem-solving over recognition. The lasting influence of architectural work—potentially affecting lives for decades—simultaneously elevates architects' sense of importance and underscores their profound responsibility, making reputation-based motivational strategies.

Vári-Szilágyi (1987) reveals that architects' identification significantly influences their attitudes, with those deeply identifying showing greater openness to collaboration, while others cling to traditional stereotypes emphasizing artistry and prestige (Caven & Diop, 2012). This explains the persistent disconnect between normative theory (idealized "good design") and positive theory (actual building performance). Such traditional attitudes resist change despite professional experience, revealing entrenched stereotypes within the field (Plebankiewicz et al., 2019).

As technology and social developments transform architectural roles, this disconnect widens (Çıdık et al., 2017). Architecture's predominant use of normative theory over positive theory, which examines actual human behaviors, directly contributes to building underperformance (Kostić et al., 2021). While behavioral sciences could integrate normative ideals with tools that accommodate human behavior (Lang, 1988), understanding what fundamentally motivates architects becomes critical to implementing such integration effectively.

2.1.2 Architects' Judgment, Decision-Making, and Behavioral Tendencies

Additionally, architectural practice embodies the tension Schön (1983) and Argyris (1982) identified between espoused theory and theory-in-use in how buildings function compared to their design intent (Question 1a). Buildings are frequently utilized differently than architects intended, due to this disconnect between designer vision and actual user needs (Question 2). Architectural design should arguably prioritize creating versatile spaces that accommodate diverse activities and social interactions over architects' visions (Baum & Valins, 2024). While architects should function as "*learning systems*" capable of transformation without losing design identity, the profession often operates in single-loop rather than double-loop learning patterns that question underlying assumptions (Argyris, 1982). The disconnect between design intent and actual building operation challenges the reflective practitioner model, as architects often create based on idealized usage rather than how occupants actually interact with spaces. This gap highlights Schön's (1983) argument that traditional analytical thinking proves insufficient in practice (Pol, 2006). Architects must develop reflection-in-action capacities responsive to post-occupancy feedback, evolving beyond designing desirable futures to "*invent ways of bringing it about*" through continuous learning cycles embracing both form and lived experience (Schön, 1983). This reflective approach is particularly relevant for encouraging positive change in the profession.

2.1.3 Mental Shortcuts (Heuristics)

Methods of certifications encourage a '*point hunting*' approach as they frequently encourage a strategic optimization approach where architects prioritize low-cost, easily attainable certifications

rather than substantial improvements (Questions 1b & 1c). For example, since energy efficiency upgrades tend to require higher financial investment compared to other certification options, projects commonly minimize these more impactful but expensive improvements in favor of more economical alternatives (Capeluto, 2022; Newsham et al., 2009). Gerstein et al.'s (1988) research on heuristics reveals why relying on certification checklists alone fails to promote sustainable behavior. This is supported by a study conducted by Acar & Yalçın (2019), which found that awareness specifically related to sustainable design practices in architecture (particularly LEED criteria) was a much stronger predictor of actual pro-environmental behavior than general environmental awareness. When architects use mental shortcuts like representativeness, availability, and anchoring instead of detailed calculations, they make systematically flawed judgments about environmental actions (Klotz et al., 2019). The multi-stage nature of decision-making, information gathering, evaluation, expression, and feedback, underscores why certification systems focused primarily on building performance miss crucial opportunities to influence human behavior.

Without addressing these cognitive processes, even certified sustainable buildings may not achieve their intended outcomes, as the complex reality of human decision-making remains unaddressed by prescriptive checklists. A study conducted by Magent et al., (2009) highlights that decision-making is central to sustainable design because these buildings require highly integrated systems that depend on well-timed, properly sequenced choices made with adequate information. When decisions are made too early without sufficient data or too late causing delays, the resulting design suffers from inefficiencies, missed opportunities for system integration, and ultimately suboptimal building performance. This highlights the necessity of examining sustainable architecture through a systems stance—one that views the challenge through multiple layers including technical, cognitive, economic, and social dimensions. The complex interplay between these factors demands analysis at different scales and perspectives if we hope to fully understand and address the limitations of current certification approaches.

2.2 Practice (Firms)

2.2.1 Certifications

Architects fundamentally shape sustainability outcomes across environmental, economic, and social dimensions through their approach to designing the built environment. Their work is crucial for advancing holistic sustainability goals (Glasson & Cozens, 2011; Lami & Mecca, 2020). Sustainable certification systems like LEED, BREEAM, WELL and LBC function as powerful behavioral drivers within the architectural profession, shaping how architects select and execute projects through credentials that serve as professional incentives (Lang, 1988). This directly addresses how incentive structures shape behavioral responses toward sustainability (Question 1b) and how the choice infrastructure of architectural practice influences design decisions (Question 1c). This behavioral influence on architects is fundamental yet often overlooked, and yet architects themselves are subject to the same cognitive biases affecting building occupants. While these systems have advanced environmental metrics, they operate on the flawed assumption that efficient buildings naturally produce sustainable behaviors, which research increasingly refutes (Jamoussi et al., 2022; Schroeder, 2018). The critical gap remains the absence of a framework connecting architectural design principles with behavioral science and occupant patterns (Hong et al., 2017).

This disconnect undermines sustainability goals, particularly with LEED, the most widely adopted system globally—which minimally addresses human factors while focusing predominantly on technical performance. Even WELL and LBC primarily address physiological wellness rather than systematically examining how buildings influence decision-making and sustainable behaviors. The profession needs assessment metrics that evaluate buildings on their demonstrated ability to facilitate sustainable behaviors over time—moving beyond prescriptive checklists toward performance-based evaluations.

The limitations of current approaches are evident in operational practices, in which meeting LEED specifications creates a significant performance gap when buildings achieve certification but fail to maintain optimal operation. The low adoption of recommissioning reflects a troubling reality that undermines long-term sustainability goals, whether from budget constraints, awareness gaps, or organizational priorities misaligned with ongoing performance verification.

2.2.2 Sustainability Trend/ Wave

Concurrently sustainability has increasingly become a marketing strategy in the built environment sector, with companies often prioritizing the appearance of sustainability over meaningful environmental improvements (Leskinen et al., 2020; Plebankiewicz et al., 2019). This trend illustrates how the framing of sustainability impacts professional practice (Question 1a), as Green certification and marketing can increase property value, incentivizing superficial sustainability practices. While some argue that even minimal sustainable adoption represents progress, such surface-level approaches can create a false sense of achievement and potentially undermine genuine environmental initiatives. This trend has prompted the industry to develop more rigorous certification systems that demand quantifiable and evidence-based sustainable design decisions.

This tendency toward superficial sustainability implementation is further complicated by fundamental definitional challenges within the field. The very malleability that makes sustainability an appealing concept across diverse contexts also creates vulnerabilities in its practical application, as the absence of standardized terminology enables varied interpretations that can serve marketing objectives rather than environmental imperatives.

2.3 *Disciplinary Norms (Architecture as a discipline)*

2.3.1 Language/Terminology

Sustainability is not a fixed concept but highly contested, context-specific, and constantly transformed (Schroeder, 2018). This conceptual fluidity directly relates to how the framing of sustainability impacts professional practice (Question 1a). The challenge of defining sustainability within the built environment stems from its inherently fluid and expansive nature (Hay et al., 2018; Kibert, 2016). Sustainability lacks a formal, internationally agreed definition, which allows the term to be used interchangeably across multiple domains, from green design and energy efficiency to material lifecycles and regenerative design principles. This linguistic ambiguity means that professionals can invoke "*sustainability*" to describe everything from net-zero buildings to biophilic design strategies, potentially diluting its precise meaning (Ding, 2008). The lack of a standardized definition creates both opportunity and confusion, as stakeholders can interpret and apply the concept flexibly, but risk miscommunication by assuming shared understanding where none truly exists.

The ambiguity in sustainability terminology creates an artificial barrier to effective implementation, as professionals operating with different interpretations cannot develop cohesive strategies or evaluate outcomes consistently. This mirrors what research has shown about ambiguity *"playing a significant role in shaping responses to stimuli by accommodating heterogeneous interpretations"* across various contexts (Duarte, 2024). Sustainability terminology shifts meaning across professional domains, creating systemic ambiguities at multiple levels that generate unintended consequences (Duarte, 2024). Without addressing this fundamental linguistic challenge, the industry risks perpetuating a fragmented approach where superficial applications of sustainability principles prevail over meaningful, measurable environmental improvements. This fragmentation not only undermines effective implementation but also overlooks potential ethical consequences, including harm to individuals, communities, and the environment itself (Guzzo et al., 2023; Guzzo & Pigosso, 2024; Lambert et al., 2019; Lorenc & Oliver, 2014). Moving forward requires acknowledging that ambiguity is not merely a problem to be *"solved"* but a complex characteristic requiring cyclical, adaptive approaches that can accommodate multiple perspectives while still driving measurable progress in promoting sustainable behaviors among occupants.

2.3.2 Occupants Behavior

It is important to understand occupant behavior in general before specifically examining sustainable behavior. Occupants are the primary consumers of energy in buildings, not the buildings themselves (Hong et al., 2017). This understanding is central to exploring how behavioral insight can help architects design built environments that encourage sustainable behavior in inhabitants (Question 2). Sustainable architecture follows a design approach driven by human behavior and needs, aiming to create a *"sustainable"* relationship between people and the built environment. It produces spaces that remain compatible with human behavior and lifestyles for extended periods of time (Kefayati & Moztarzadeh, 2015). The architect's role is crucial, as sustainable spaces require intentional and proper design guided by specific values to positively affect the people who inhabit them (Eizenberg & Jabareen, 2017; Sodagar et al., 2008). Occupant behavior emerges as a critical yet unpredictable factor in building energy consumption (Ahn et al., 2017; Yan et al., 2015). The significant impact of human behavioral patterns introduces substantial uncertainty in energy performance predictions, highlighting the urgent need for more sophisticated occupant behavior modeling methodologies (Schweiker & Wagner, 2016). While recognizing the crucial role of understanding human interactions with built environments, researchers face ongoing challenges in developing tools that can effectively capture the complex and inherently variable nature of occupant behaviors.

Alternatively sustainable building design often fails to promote sustainable occupant behavior due to architects' overemphasis on technical solutions while neglecting how people actually use buildings (Yousefi et al., 2017). The environmental performance of a building critically depends on occupant behavior, with many sustainable designs underperforming because users override smart systems, ignore recycling areas, or disable energy-efficient features. Rather than expecting occupants to adapt to idealized designs, architects can benefit from integrating behavioral science to further understand, inform users (raise awareness), and create intuitive, user-friendly sustainable features that naturally encourage environmentally responsible choices.

2.3.3 Quantitative Bias/ Hard Science Drive

Traditional distinctions between "hard" natural sciences and "soft" social sciences are becoming less relevant in contemporary problem-solving (Shapin, 2022). Yet architectural practice continues to privilege quantifiable approaches over behavioral understanding (Question 1c). The systematic neglect of behavioral science principles in architectural practice undermines sustainability efforts and stems from architects' professional motivations, which prioritize demonstrable metrics over actual performance outcomes. While fields converge to address complex challenges, architecture remains tethered to measurement systems that often fail to capture the human dimensions essential to true sustainability.

Architects' behavior is driven by external standards like LEED certification and internal identity-based indicators such as professional reputation. Schön (1983) identified this problem as an overreliance on "technical-rationality," arguing instead for "an alternative epistemology of practice" that recognizes how professional knowledge is situational and contextual rather than universal and abstract (Carmon & Mannheim, 1979). This creates a cycle where the bias toward quantifiable metrics manifests in professional behaviors that prioritize what is easily measured over what might be most effective.

The consequences of this bias are evident in the well-documented "*energy performance gap*" between theoretical calculations and actual building performance (De Wilde, 2014; Turner & Frankel, 2008; Norford et al., 1994). Newsham et al. (2009) revealed that 28-35% of LEED-certified commercial buildings actually consumed more energy than their non-certified counterparts, with no significant correlation between certification levels and actual performance. Yet these certifications continue to drive architectural practice because they offer quantifiable validation that shapes professional identity, despite researchers including Amiri et al. (2019) and Scofield (2013) finding little relationship between certification levels and actual energy outcomes.

The compelling impact of occupant behavior—which can vary energy consumption by up to 150% even with identical building designs (Yan et al., 2015)—remains understudied precisely because it resists easy quantification. Researchers often favor reliable numerical approaches despite the importance of social-psychological factors (Hong et al., 2017; Uddin et al., 2021), perpetuating a professional bias toward what can be measured rather than what should be addressed.

Current research gaps that I am highlighting are—insufficient integration of behavioral theories with qualitative measurements, inadequate interdisciplinary collaboration, and limited exploration of alternative frameworks (Heydarian et al., 2020)—persist not because they are unrecognized but because addressing them requires architects to venture beyond the quantifiable metrics that currently define professional success. This disconnect between design intentions and actual usage patterns demonstrates that truly sustainable architectural design requires equal attention to human behaviors lens from a practice and end-user stand that ultimately determine resource consumption and environmental impact, not just "*hard science*" engineering solutions that can be easily quantified, measured and certified.

The empirical context outlined above reveals several critical gaps in current architectural practice related to sustainability. The overemphasis on quantifiable metrics at the expense of behavioral understanding, the disconnect between certification systems and actual performance, and the

contested nature of sustainability terminology all point to the need for a methodological approach that can bridge these divides. The following mixed-methods research design directly addresses these gaps by examining how framing, incentives, and choice infrastructure function as interconnected systems that shape sustainable outcomes in architectural practice. This approach deliberately spans multiple analytical levels, from individual architect decision-making to organizational processes to disciplinary norms, to capture the complex behavioral systems that determine sustainable architectural outcomes.

Section 3 Principles & Methodology

Mixed Methods Approach & Triangulation

This research employs a mixed-methods approach to gain comprehensive insights into architectural practice through the lens of the behavioral systems design. This integrated methodology combines quantitative and qualitative techniques to examine the relationship between framing, infrastructure, incentives, and behavioral outcomes in sustainable architectural practice. The mixed-methods design was selected to provide triangulation across data sources, strengthening the validity and reliability of findings while capturing the complexity of behavioral systems design in architectural practice. Triangulation serves as a critical methodological strategy in this research, strengthening validity and enhancing the depth of understanding across multiple dimensions. By systematically combining diverse data collection methods and sources, this study examines behavioral systems in architectural practice from complementary perspectives - analyzing published discourse (systematic literature), mapping organizational processes, documenting practitioner experiences, and evaluating built outcomes. This multi-layered triangulation approach not only validates findings through convergence but also illuminates richer insights through divergence, revealing tensions between how sustainability is framed, how practice infrastructure shapes decisions, how incentives drive behavior, and how these elements manifest in built environments.

3.1 Question 1- How can architectural practice use a behavioral systems design lens to inform more sustainable built environment/ architectural design practice?

Systematic Analysis & Semi- Structured Interviews

3.1.1 Question 1a: How does the framing of "sustainability" impact professional practice?

The research employs discourse systematic analysis of architectural publications to examine how sustainability is conceptualized within the profession. This analysis encompasses influential articles from leading architectural journals, sustainability statements from diverse architectural firms, certification selection and criteria, and key professional guidance documents from industry organizations. Using qualitative data analysis software, texts will be coded to identify dominant metaphors, narratives, and conceptual frameworks around sustainability, with particular attention to how sustainability is framed at different system levels according to 3x3 Behavioral Systems Model (Fig.1 – Section 4). To complement this textual analysis, semi-structured interviews will be conducted with architects representing different roles, career stages, and across multiple firms. These interviews will explore how practitioners conceptualize sustainability, perceive tensions between different sustainability frameworks, and translate sustainability concepts into design decisions. Interview data will be analyzed using thematic analysis techniques, with resulting themes mapped against the ideological layer of the framework to identify how mental models shape practice across system levels.

3.1.2 Question 1b: How do incentive structures shape behavioral responses toward sustainability in architectural practice?

The research employs a case study approach examining architectural firms with different incentive structures related to sustainability. Firms (commercial buildings/projects) will be selected to represent diversity in how they incentivize sustainable design, including those with explicit

sustainability metrics, those structured around sustainability as a core value proposition, and conventional firms without specific sustainability incentives. For each case, data collection will involve reviewing firm policies and communications, interviewing members at different organizational levels, and analyzing recent sustainability-focused projects. These case studies will examine how formal and informal incentives align or conflict with sustainability goals. To gather broader perspective on incentive structures, an online survey will be administered to architects beyond the case study firms. This survey will explore perceptions of various incentive structures, their prevalence, and their perceived effectiveness in promoting sustainable design, providing valuable context about incentive landscapes in contemporary architectural practice.

Ethnography & Survey

3.1.3 Question 1b: In what ways does 'choice infrastructure' of architectural practice shape design and development decisions?

My research focuses on two interconnected elements of choice infrastructure that fundamentally shape sustainable design decisions in architectural practice. First, I'm examining project delivery methods and organizational structures that establish the frameworks determining when and how sustainability considerations enter the design process, including contract structures, stakeholder engagement timing, and decision hierarchies within firms. Second, I'm investigating internal review and decision processes such as stage-gate reviews, design critique methodologies, and quality assurance protocols that influence how sustainability factors are weighted against competing priorities. These choice infrastructural components work together as an interconnected system that shapes architects' decision-making about sustainability without directly dictating specific design outcomes, creating both constraints and opportunities for sustainable practice.

To explore these two interconnected elements of choice infrastructure, I will be an ethnographic observer study (inside out view/perspective) within a number of architectural firms in Chicago, complemented by critical decision method analysis. Through participant observation, I will be immersed within selected architectural practices during active commercial sustainable design projects, documenting how project delivery methods and organizational structures interact with internal review processes as sustainability decisions naturally unfold. This ethnographic approach reveals both formal and informal aspects of the firm's choice infrastructure by shadowing key team members, attending design meetings, client presentations, and internal reviews where sustainability decisions are negotiated and finalized. For example, one of the firms I would like to look into is Gensler. I would observe how their matrix organizational structure, with sustainability specialists distributed across different project types and market sectors, influences when and how sustainable design strategies are introduced into projects. I will document how their established design review process shapes decision-making at different project stages, revealing how sustainability considerations compete with other priorities during critical design reviews. My observations will capture visible decision pathways, information flows between stakeholders, and critical moments where sustainability ambitions confront practical limitations. For deeper insight into specific decisions, I will conduct targeted follow-up interviews to investigate how organizational structures and review processes influenced pivotal sustainability decisions. Special attention will be paid to identifying tacit knowledge, unwritten rules, and organizational habits that shape sustainability outcomes but may not appear in formal process documentation. This methodology provides an "inside out view" of architectural decision-making rather than merely

capturing idealized processes, revealing how choice infrastructure functions as a living system within architectural practice.

Following this qualitative phase, I will develop and distribute a survey to a broader sample of architectural practitioners across various firm types and sizes. This survey will test whether the patterns and mechanisms identified in the ethnographic study are consistent across the profession, while also gathering quantitative data on how different organizational structures and review processes correlate with sustainability outcomes. Allowing me to develop rich, contextual understanding through ethnography while also establishing the prevalence and significance of these choice infrastructure elements across architectural practice more broadly.

Case Study: Kaplan Building

Question 2: How can behavioral insight help architects design-built environments that encourage sustainable behavior?

The final phase of my research employs a detailed case study approach centered on the Kaplan Institute Building. This building analysis serves as a practical application that connects theoretical insights from my investigation of architectural practice to tangible built outcomes. The Kaplan Building functions essentially as a living laboratory where I can observe how the behavioral systems uncovered in Question 1 (framing, incentives, and choice infrastructure) materialize in physical form. My methodological approach integrates findings from earlier research phases to guide this investigation. Data collection methods include comprehensive review of design documentation, semi-structured interviews with the John Ronan Architects team, on-site structured observations of building usage patterns, and analysis of performance metrics. During architect interviews, I will specifically probe how sustainability framing concepts influenced design decisions, how project incentives shaped feature selection, and how their organizational processes facilitated or constrained sustainable solutions. This real-world testing environment allows me to trace connections between practice systems identified in Question 1 and their manifestation in built form.

The user perspective forms another critical dimension of this study. I will administer targeted surveys to Kaplan Building occupants, supplemented by follow-up interviews with select users. These instruments will assess occupants' awareness of sustainability features, document interaction patterns, measure perceived behavioral influences, and evaluate overall satisfaction. The survey design will deliberately test whether occupants experience the building in ways that reflect the framing concepts, incentive structures, and choice infrastructure identified through my practice research. This approach establishes a valuable feedback loop between my research questions. Findings about architectural practice systems inform my analysis of the Kaplan Building, while observations of the building simultaneously validate or challenge my understanding of practice systems. By tracing how behavioral design principles transition from practice methodology to built reality, my research aims to develop a holistic understanding of sustainable architectural practice as an integrated behavioral system that spans from initial concept development through actual building occupation.

Section 4 Design Research Framework

4.1 Systems/Levels

A systems lens positions behavioral design research as a powerful connector bridging human perspectives across micro (individual), meso (group), and macro (societal) levels. Examining phenomena across all three levels reveals intricate interactions that remain obscured when studying each level in isolation (Meadows, 2008; Cicchetti & Dawson, 2002). This multi-dimensional perspective shows how individual components, organizational structures, and broader systemic influences dynamically interact and co-evolve. By bridging disciplinary boundaries and offering a holistic view, the systems approach develops more nuanced insights and effective intervention strategies across complex domains (Cabrera & Cabrera, 2023). Conceptualizing levels as leverage points and viewing interventions as interconnected events enables designers to create sophisticated solutions addressing intricate challenges with greater systemic awareness (Maier & Cash, 2022). Complex societal problems require systemic thinking where design serves as the integrative discipline guiding system evolution, coordinating short-term interventions with long-term goals. This approach provides deeper comprehension of how micro-level behaviors connect to meso-level organizational dynamics and macro-level systemic influences, yielding a more comprehensive understanding of this complex phenomena.

4.2 Framework: Using a Behavioral System Design Lens into Architectural Practice

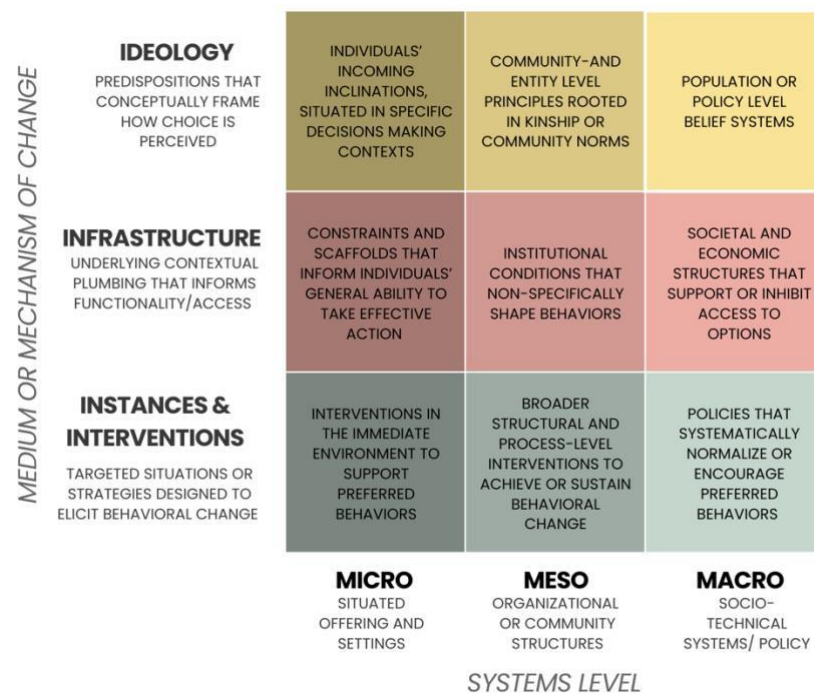


Fig. 1 – 3x3 Mapping system conditions for behavioral change by scale and mechanism explicitness/embeddedness by Ruth Schmidt (Schmidt, 2023)

The research questions in this study operate within a behavioral design system framework, which offers a powerful structure for examining how architectural practice influences sustainable outcomes. My primary research question—"How can architectural practice use a behavioral systems lens to inform more sustainable built environment/architectural design practice?"—positions this investigation at the intersection of behavioral science and architectural practice, recognizing that architecture functions as a complex system. The sub-questions deliberately explore different aspects of this framework to ensure comprehensive exploration.

The first sub-question, "How does the framing of 'sustainability' impact professional practice?" examines the ideological dimension across system levels, investigating how conceptual understandings of sustainability shape architectural thinking and decision-making. The second sub-question, "In what ways does 'choice infrastructure' of architectural practice itself shape decisions about design and development?" focuses on the infrastructure-meso intersection, analyzing how organizational structures and procedural systems influence sustainability decisions within firms. The third sub-question, "How do incentive structures shape behavioral responses toward sustainability in architectural practice?" investigates intervention mechanisms, examining how motivational factors drive sustainable design choices.

The second primary research question, "How can behavioral insight help architects design built-environment-as-a-system to encourage sustainable behavior in inhabitants?" extends the investigation to the outcomes of architectural practice. This question operates primarily across the intervention dimension of the framework at all system levels (micro, meso, and macro), while also acknowledging connections to infrastructure and ideology. At the micro level, it examines how specific design elements directly influence individual behaviors. At the meso level, it explores how building systems and organizational spaces shape collective patterns of interaction. At the macro level, it investigates how architectural interventions within broader urban contexts can promote sustainable behaviors at community scale. By framing the built environment as a system rather than a static artifact, this question acknowledges the dynamic interactions between spatial configurations and human behavior patterns, recognizing that buildings themselves function as intervention mechanisms within larger social and environmental systems.

This framework enables me to locate where current architectural approaches are inadequate. While architecture has adopted some behavioral tools at the intervention level and created infrastructure at the meso level, the profession lacks comprehensive integration of behavioral science across all matrix intersections. This creates a research opportunity to develop strategies for architectural practice that function effectively across multiple system levels and mechanisms of influence. The framework helps identify strategic leverage points for my research. Addressing ideology at the micro level could create cascade effects throughout the system, while creating new choice infrastructures at the meso level might help bridge the gap between design intent and actual behavioral outcomes. The framework reveals why previous attempts to integrate behavioral insights into architecture have fallen short, as many efforts have focused on single cells of the framework without addressing complex interactions between system levels and mechanisms.

Fundamentally, this approach provides a structured way to organize my research methodology and findings, ensuring the research addresses the full complexity of architectural practice rather than focusing on isolated components. It positions my work to make a significant contribution by

offering a holistic behavioral systems design model to inform architectural practice, moving beyond simplistic interventions to address the fundamental systems that shape how architecture is practiced and experienced.

This research contributes to both architectural practice and behavioral design by developing a comprehensive framework that bridges historically disconnected domains. By examining how behavioral design systems operate across multiple dimensions of architectural practice—from individual architects' decision-making processes to firm-level infrastructures to profession-wide framing of sustainability—this study promises to identify critical leverage points for intervention. The multi-method approach ensures triangulation between textual, observational, and experiential data, while the case study component grounds theoretical insights in practical application. As architectural practice confronts increasing pressure to address environmental challenges, this behavioral design systems approach offers a pathway to move beyond superficial sustainability metrics toward design approaches that genuinely support sustainable outcomes and human/occupant behavior.

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